Representing Code

Introduction

At this point we have source code

which turn into

tokens

next is to use those tokens to make something the parser likes more

Representing code

our representation should be easy for the **parser** to produce and easy for the **interpreter** to consume

we can figure out what a good representation is by looking at how a *human* interprets

```
1  1 + 2 * 3 - 4
```

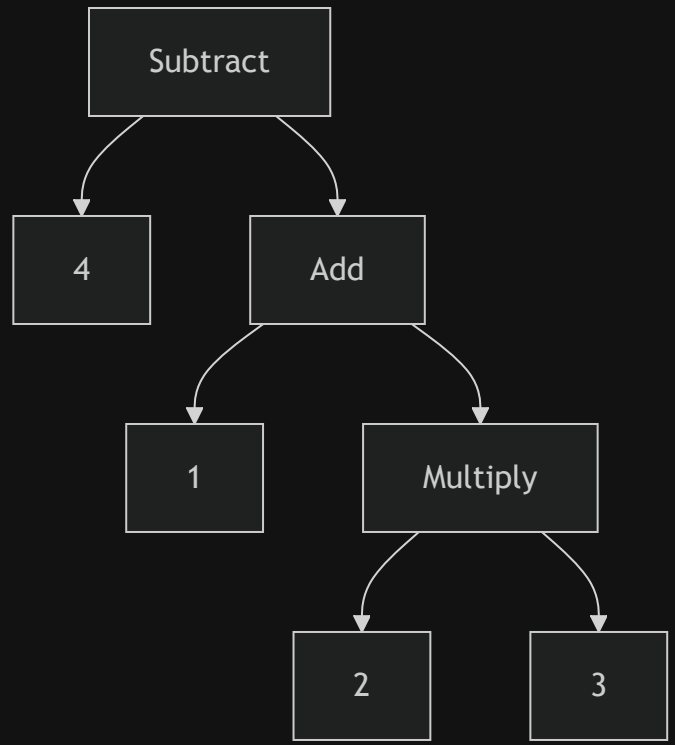
Representing code

Because of grade school, we know that we do multiplication first

One way to represent this is to use a tree structure

In order to evaluate a node of arithmetic, we need to know the literal values of the leaf nodes, so we evaluate that first

So starting from the leaves, up to the root (post-order traversal)



Exercise

Draw the syntax tree for the following expression:

```
(5 - 2) * (8 + 3)
2 * 4 + 6 / 3
10 - (2 + 3) / 4
```

Context free grammars

Defining our lexical grammar, the rules for how characters get grouped, the grammar for lexemes, was called a **regular grammar**

And that works for scanners which produce a flat sequence of tokens

Because we need to do nesting, we need something more powerful

in the toolbox of formal grammar, this is called a **context free grammar**

Context free grammars

A formal grammar takes a set of atomic pieces which it calls "alphabet", then it defines a set of "strings" that are "in" the grammar, and each string is a sequence of "letters"

In our scanner,

- the alphabet is characters
- the strings are valid lexemes

in our parser

- the alphabet is tokens
- the strings are valid expressions

Notation

> how do we write down a grammar that contains an infinite number of valid strings?

It's easier to define the garmmar using rules

Starting from the rules you can *generat* strings that are in the grammar, this is called derivations

And so rules are called *productions* because they produce srtings in the grammar

Notation

A context free grammar has a

- head, which is the name
- a body, which describes what it generates

and a body has a list of symbols of two types

- a terminal, from the grammars alphabet, which are end points
- a non-terminal, which is a reference to another production

And finally, you can have rules with the same name, and when you hit a non terminal with that name, you can pick any of the rules

Notation

To write this down, we have a variety of proper notations, we'll be using a modified Backus-Naur Form (BNF)

but ways to crystalize grammars have been invented as far back as tthe ashtadhyayi (600 BCE)

each rule is a name, followed by an arrow (->), a sequence of symbols, and a semicolon (;)

teriminals are quoted strings, non terminals are lowercase words

Example

```
breakfast → protein "with" breakfast "on the side";
breakfast → protein;
breakfast → bread;

protein → crispiness "crispy" "bacon";
protein → "sausage";
protein → cooked "eggs";

crispiness → "really";
crispiness → "really" crispiness;

cooked → "scrambled";
cooked → "fried";

bread → "toast";
bread → "bagel";
```

Exercise

Using the grammar from the previous slide, generate some sentences

Neo

Enhancing the notation

Purely to make it easier to read and write, we can add some enhancements to the notation

- the pipe (|) means "or", so we can combine multiple rules with the same name
- parenthesis () group things together, meaning to select from one of the options inside
- the asterisk (*) means "zero or more" of the preceding item
- the plus (+) means "one or more" of the preceding item
- the question mark (?) means "zero or one" of the preceding item

```
1  breakfast → protein ( "with" breakfast "on the side")? |
2              bread;
3
4  protein →  "really"* "crispy" "bacon" |
5              "sausage" |
6              ("scrambled" | "fried") "eggs";
7
8  bread →    "toast" |
9              "bagel";
```

Making a grammar for (a portion of) lox

Some definitions

The syntactic grammar is significantly larger than the lexical grammar, so we'll only show a portion of it here

the main things we want to parse are

- literals: numbers, strings, booleans, nil
- unarys: prefixes like (!) and (-)
- binarys: infix like (+), (*), etc, and logical operators (==, <=)
- parentheses: a pair of ( ) to group expressions

Which lets us run things like

```
1   1 - (2 * 3) < 4 == false
```

Class Exercise

Finish the grammar

```
expression → literal | unary | binary | grouping;

literal → NUMERAL | STRING |
grouping →
unary →
binary →
operator →
```

so that

```
1 - (2 * 3) < 4 == false
```

is a valid expression

Implementation of Syntax Trees

You'll notice that in a grammar, almost all productions refer back to "expression"

This makes a tree structure a natural fit for representing code

> what is a tree for syntax called?

A single expression class with a list of children would also work

We'll mainly implement them as individual classes since we're using java

So for each production under expression, we make a subclass that has fields for the non-terminals specific to that production

Example

```
abstract class Expr{
    static class Binary extends Expr {
        Binary(Expr left, Token operator, Expr right) {
            this.left = left;
            this.operator = operator;
            this.right = right;
        }
        final Expr left;
        final Token operator;
        final Expr right;
}
```

Where `Expr` is a base class that all expressions inherit from

> all operations in our language will be considered an expression

Metaprogramming

Since all we're doing is making behavior less classes,

We can use **metaprogramming** to generate the classes for us

> When given a task that is mostly boilerplate, write code to write that code for you

Metaprogramming

```
tool/GenerateAst.java
create new file
```

```java
package com.craftinginterpreters.lox.tool;

import java.io.IOException;
import java.io.PrintWriter;
import java.util.Arrays;
import java.util.List;

public class GenerateAst {
    public static void main(String[] args) throws IOException {
        if (args.length != 1) {
            System.err.println("Usage: generate_ast <output directory>");
            System.exit(64);
        }
        String outputDir = args[0];
    }
}
```

Defining the AST structure

```
tool/GenerateAst.java
in main()
```

```
        // String outputDir = args[0];
        defineAst(outputDir, "Expr", Arrays.asList(
            "Binary   : Expr left, Token operator, Expr right",
            "Grouping : Expr expression",
            "Literal  : Object value",
            "Unary    : Token operator, Expr right"
        ));
```

Defining the AST structure

```
tool/GenerateAst.java
add after main
```

```
private static void defineAst(
    String outputDir, String baseName, List<String> types)
    throws IOException {

    String path = outputDir + "/" + baseName + ".java";
    PrintWriter writer = new PrintWriter(path, "UTF-8");

    writer.println("package com.craftinginterpreters.lox;");
    writer.println();
    writer.println("import java.util.List;");
    writer.println();
    writer.println("abstract class " + baseName + " {");

    writer.println("}")
    writer.close();
}
```

Defining the AST structure

```
tool/GenerateAst.java
in defineAst(), before writer.close()
```

```
    // writer.println("abstract class " + baseName + " {");
    for (String type : types) {
        String className = type.split(":")[0].trim();
        String fields = type.split(":")[1].trim();
        defineType(writer, baseName, className, fields);
    }
```

Defining type

```
tool/GenerateAst.java
add after defineAst()
```

```
private static void defineType(
    PrintWriter writer, String baseName,
    String className, String fieldList) {
    writer.println("    static class " + className + " extends " + baseName + " {");

    // Constructor.
    writer.println("        " + className + "(" + fieldList + ") {");

    // Store parameters in fields.
    String[] fields = fieldList.split(", ");
    for (String field : fields) {
        String name = field.split(" ")[1];
        writer.println("            this." + name + " = " + name + ";");
    }
    writer.println("        }");
```

Defining type

```
tool/GenerateAst.java
add after constructor
```

```
        // Fields.
        for (String field : fields) {
            writer.println("        final " + field + ";");
        }
        writer.println("    }");
    }
```

If you compile and run this code, it will generate the Expr.java file for us

Working with trees

Imagine what the interpreter will do

> The interpreter needs to do different things depending on what kind of expression it is evaluating

Unlike with tokens, where we can just `switch` on the token type

We can’t have a "type" for each expression, because each expression is a different class

Technically

We can simply

```
if (expr instanceof Expr.Binary) {
    // handle binary
} else if (expr instanceof Expr.Literal) {
    // handle literal
} ...
```

But this is slow and clunky

Since we have an object-oriented language, we would want to put behavior into each class

Simply having an `intepret()` method on the `Expr` superclass that each subclass overrides

> However, this scales poorly

Both the parser and the interpreter would need to be modified every time we add a new expression type

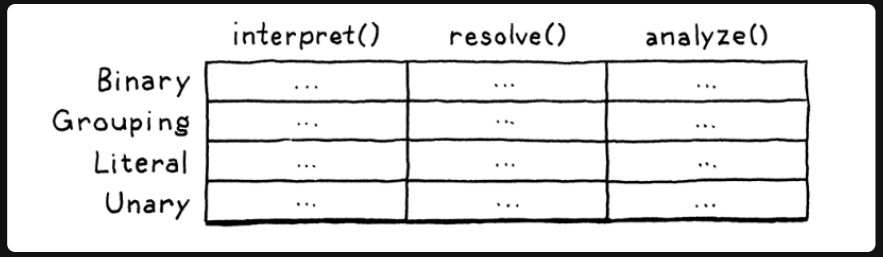
The Expression Problem

The expression problem

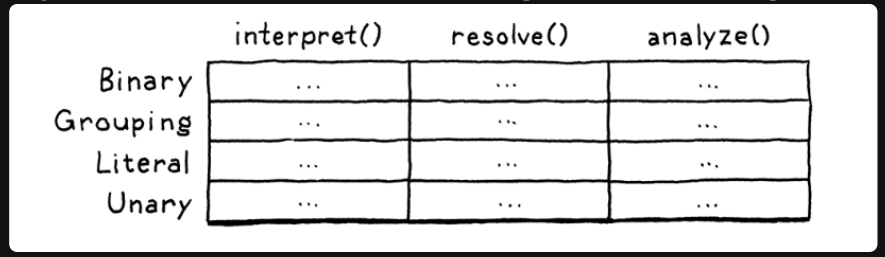
We have some types (binary, literal, grouping, unary)

and a few operations (intepreter, resolve, analyze)

And for each pair we need an implementation



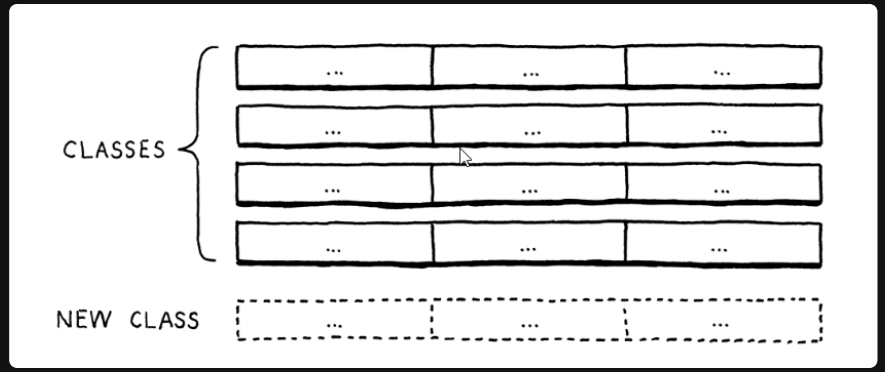
The nature of Object Oriented Programming



Object oriented programming likes writing new rows

> the core idea is that the things you do with a type are likely related

So it tries to make it easy to define them together as methods inside a class

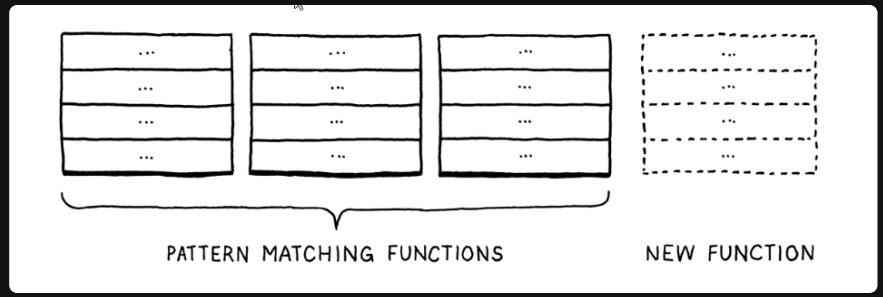


The nature of Functional Programming

In functional programming, there exists methods, not classes.

> Types and functions are distinct

To implement an operation for a number of different types, you make **one** function that pattern matches on the type



So to make a new operation, you just make a new function, and implement all of its behaviors depending on the type

Problem

There's a "*grain*" to each paradigm

And it's difficult to do one thing in the style of the other

This is called the **expression problem**

The Visitor Pattern

The visitor pattern

A pattern about *approximating* a functional style within OOP

Its goal is allowing us to make a new set of columns easier

We simply define all the behavior for a new operation in one place

An Example

```
1  abstract class Pastry {}
2
3  class Beignet extends Pastry {}
4
5  class Cruller extends Pastry {}
```

And we want to define new pastry operations `cooking, eating, decorating, etc`

And we want to do that without having to add a new method to each class every time

An Example

To do that we make an **interface**

> in java, an interface is a class that only has method signatures, no implementations

Like a blueprint for classes, or a class for classes

```
1  interface PastryVisitor {
2      void visitBeignet(Beignet beignet);
3      void visitCruller(Cruller cruller);
4  }
```

Where each operation that we want to do on pastries is a new class that **implements** the interface

So now we have classes for the **objects** (pastries) and classes for the **operations** (visitors)

An Example

Then we simply route those two using some *polymorphism*

```
abstract class Pastry {
    abstract void accept(PastryVisitor visitor);
}
```

And each subclass implements

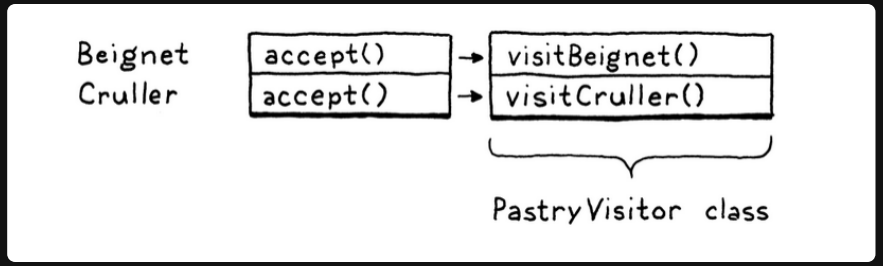
```
class Beignet extends Pastry {
    @Override
    void accept(PastryVisitor visitor) {
        visitor.visitBeignet(this);
    }
}
```

```
class Cruller extends Pastry {
    @Override
    void accept(PastryVisitor visitor) {
        visitor.visitCruller(this);
    }
}
```

An Example

Then finally, to perform an operation on the pastry,

we simply run the `accept()` method with the visitor just passed in



This means all we need to do to add a new operation is make a new class that implements the visitor interface

Like a `CookingVisitor`, `EatingVisitor,` etc, and then pass that to the pastry's `accept()` method

Visitor Pattern for Expressions

We'll be defining the visitors inside the generate ast class so we don't have to write them by hand

```
tool/GenerateAst.java
in defineAst()
```

```
    // writer.println("absract class " + baseName + " {");
    defineVisitor(writer, baseName, types);
```

Visitor Pattern for Expressions

```
tool/GenerateAst.java
add after defineAst()
```

```
private static void defineVisitor(
    PrintWriter writer, String baseName, List<String> types) {

    writer.println("    interface Visitor<R> {");

    for (String type : types) {
        String typeName = type.split(":")[0].trim();
        writer.println("        R visit" + typeName + baseName + "(" +
            typeName + " " + baseName.toLowerCase() + ");");
    }

    writer.println("    }");
}
```

Visitor Pattern for Expressions

```
tool/GenerateAst.java
in defineAst()
```

```
    // defineType(writer, baseName, className, fields);
    // }

    writer.println();
    writer.println("    abstract <R> R accept(Visitor<R> visitor);");

    // writer.println("}");
```

Visitor Pattern for Expressions

Finally, each subclass should implement the accept function

```
tool/GenerateAst.java
in defineType()
```

```
    // writer.println("    }");

    // visitor pattern
    writer.println();
    writer.println("        @Override");
    writer.println("        <R> R accept(Visitor<R> visitor) {");
    writer.println("            return visitor.visit" + className + baseName + "(this);");
    writer.println("        }");
    writer.println("    }");

    // fields
```